



Crowdsourced Project Selection in Tourist Spots with Voting in Smart Cities

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Abstract. The tourism industry is a valuable sector generating major income for nations. The local Government of a tourist spot may undertake several projects to be implemented for the benefit of tourism in that place. Currently, the authority of the local Government, involving their ministers, decides which projects are to be undertaken. Alternatively, the local Government may involve, in a crowdsourced (crowdsourcing can be implemented with smart devices such as mobile devices also) manner with the technological support of smart cities, the local residents and the tourists in this process, leading to a more transparent and more engaging selection of projects. In this paper, a strategyproof, Pareto optimal, and scalable mechanism is provided to gather the preferences of the tourists, the residents and the ministers (with tunable weight parameters) for their most preferable choices of projects in the tourist spot and selecting the projects in such a manner so that the total cost of the selected projects is within the available budget.

Keywords: Tourism · Project selection · Crowdsourcing · Minister votes · Budget feasibility · Weighted votes

1 Introduction

Tourism is one of the major sources of income for many countries, if not the primary one [1]. Governments are particularly eager to take steps and create policies that promote the tourism industry. To promote tourism, (local) Government of a tourist spot may try to provide some more facilities to the tourists by undertaking several projects. To do the same, certain decisions on which projects to be undertaken in tourist spots have to be taken from time to time. These decisions, the result of which is to be increased tourism, were traditionally taken by

the Government itself involving some personnel internally (henceforth these personnel will be termed as ministers). Recently, there has been a change of trends by involving the local residents in the decision making process [2]. However, tourists are the best people to contact when it comes to deciding on the requirements of improving tourism [3,4]. Meanwhile, crowdsourcing/crowdsensing is gaining popularity throughout the world [5,6]. The distribution of tasks into the masses using minimal mobile hardware support is a great way to monitor and manage systems remotely [7]. The latest trend by the Government is the involvement of tourists and residents in the decision making process through crowdsourcing. In [2], a crowdsourcing based (i.e. involving the local residents) game theoretic framework is proposed to select a spot for tourism. This is made feasible by the technological support that is available in existing and upcoming smart cities.

However, in a tourist spot, if several projects are to be implemented by involving the local residents, ministers and tourists are not discussed in [2]. In this paper, we present a novel mechanism to combine the preference lists of the tourists, the residents and the ministers to select a list of projects that fits the budget of the Government. We take care of the situation of the votes of the ministers having a lower effect on the selection process by assigning an adjustable weight parameter as a multiplier of the ministers' vote counts. This parameter can be adjusted as per the requirements of the land to make the mechanism suitable for that project selection settings. Figure 1 presents a schematic representation of the proposed mechanism.

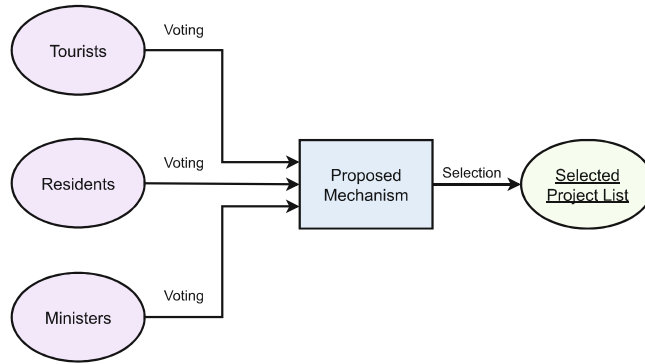


Fig. 1. A schematic representation of the proposed mechanism.

Following is the organization of this paper. Section 2 presents the related work. The problem has been formulated and the proposed mechanism has been suggested in Sect. 3. Analysis of the proposed mechanism has been done in Sect. 4 followed by its simulation results in Sect. 5. Discussion on the results is presented in Sect. 6. Conclusion and future work are put forward in Sect. 7.

2 Related Work

The involvement of local residents in Government decision making, in terms of deciding on tourist spots, has been explored [2]. Participatory democracy is an emerging field of research where participation of the people of a particular region in decision making is explored by the Governments or some organisations [8]. The involvement of tourists themselves in the development of tourism programs creates more competitive markets for the development of smart tourism projects [3, 4]. Mobile games were designed for tourists and stakeholders aimed at improving the tourism experience by optimizing services for the tourists and enhancing the understanding and belief systems among stakeholders for sustainable development of the tourism industry itself [9, 10]. Sustainable tourism development is deeply supported by the community of the land and is determined by the residents' perception of value and collaboration of the domain [11]. While mostly residents contribute without any incentives, their participation in tourism projects for its perceived benefits significantly modify their attitude towards the industry tipping the scales in favour of its development [12]. The difference in approaches of various stakeholders towards the tourism industry at different levels of action is striking [13]. Upcoming smart cities now receive more tourism than ever leading to a sustainable development of the industry there [14]. Residents protest against the development of the tourism industry in spite of the existence of proper participatory channels and encouragements provided by the Government [15].

The inclusion of the Internet of Things (IoT) as the task executors in a crowdsourcing environment has been studied as well [16]. Data labelling using crowdsourcing is a low cost way to create datasets for the training of machine learning algorithms used for a variety of purposes [17]. Geographic data collection using crowdsensing is another well researched domain [7]. Crowdsourcing has also been used to go beyond borders to multiple cities for the advertisement of tourist spots [5]. A strategical system for the reporting of data and the incentive awarded to the reporting agent while preserving privacy of the reporting agent has been suggested [6]. The selection of volunteers based on whether the volunteers have the qualities which are the fitting criteria of the task is explored [18]. Crowdsourced data has been used to address the issue of meaningful and efficient tourist guidance for relevant services [19]. Utilizing the power of the mobile devices of the masses in the operation of varied applications has been discussed [20]. Mobile crowdsourcing as an useful medium for the full utilization of the sensor-rich smartphones, has become a popular field of exploration [21, 22]. Mobile crowdsourcing solutions enable people to earn small amounts of money in exchange of the completion of simple tasks prove the efficacy of the approach [23, 24]. Not only the existing smart cities, but also the developing smart cities can benefit from the mobile crowdsourcing facilities [25]. Mobile crowdsourcing based road safety services provide real-time traffic offense reporting and mapping features [26]. Mobile crowdsourcing successfully forecast electricity loads in dynamic environments [27]. Mobile crowdsourcing is

advocated for indoor positioning systems where GPS services are not reliable or available [28].

Crowdsourcing can be applied in Intelligent Transportation Systems (ITS) as well [29]. The revolutionary idea of crowdsourcing solutions for the plethora of open challenges of the time has been explored by several organizations [30]. Time dependent critical tasks that require the task requester to accept trade-offs in terms of cost and quality have been explored [31]. The contribution of citizens in sustainable development of the society is studied [32]. Pricing policies and randomized incentive mechanisms make the approach practical and realistic for several application domains [33]. A survey of several studies spanning over three decades that explore the attitude of the local residents for a sustainable tourism has been done [34,35]. Tracking the quality of air with the help of low-power and low-cost mobile hardware and crowdsourcing has been explored [36]. Detailed research on the participation of citizens in the decision making process of the nation has been studied over several years [37]. Community resilience in the management of natural resources by the community is studied and their main challenges have been pointed out [38].

However, the aforementioned works have not designed and leveraged a proper setup to utilize the power of crowdsourcing in engaging tourists, residents and ministers together in an unified attempt to select projects for a tourist spot. In addition, these works consider all votes as equal and do not provide any facility for different weights for different stakeholders. Our work is aimed to provide a platform to select projects for tourist spots by involving tourists, residents and ministers in the process with variable weights for the votes, thereby handling both the aforementioned issues.

3 Algorithms and Methodology

The selection of projects by the local residents, the optimization of tourist routes by the tourists themselves, the predictive automation of relevant services for the tourists by using polling mechanisms, and other similar ideas are already existing and have been studied well. However, considering the ministers' preferences and employing variable weights for the votes is a novel attempt towards resolving the same challenge from the bottom up. We discuss this notion in detail here.

3.1 Notations and Problem Formulation

In this model, there are $P = \{p_1, p_2, \dots, p_q\}$ number of possible projects that are decided to be carried out. Each project has a cost associated with it. Let us denote the cost of each project by α_l and the cost vector is denoted by $\alpha = \{\alpha_1, \alpha_2, \dots, \alpha_q\}$. An overall budget Δ is sanctioned to carry out all the projects selected through the votes of the n number of tourists and residents taken together, and the m number of ministers denoted as $H = \{h_1^M, h_2^M, \dots, h_m^M, h_1^T, h_2^T, \dots, h_n^T\}$. Here $\alpha_l \leq \Delta$ and let us denote the cost of each project $p_l \in P$ by $c(p_l)$. So, by definition, $c(p_l) = \alpha_l$. Similarly, let the weight of a vote be

denoted by $\omega \in \mathbb{R}^+$. We assume a normalized value of the weights such that all tourists and residents are assigned unit weights, and all ministers are assigned a *minimum* weight of unity. So, we have $\omega_i^T = 1$ for h_i^T , and $\omega_j^M = \omega \geq 1$ for h_j^M , where $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m$. The k number of selected projects will be denoted by $S = \{s_1, s_2, \dots, s_k\}$. As the selected projects' cost will not exceed the overall budget Δ , we can write $\sum_{l=1}^k c(s_l) \leq \Delta$.

A tourist, a resident or a minister (hereafter, referred interchangeably as agents) can provide a choice of a subset of the projects from P and we can denote that by P_i . Say, for example $P = \{p_1, \dots, p_{10}\}$ and the i^{th} agent puts her choice as $P_i = \{p_1, p_4, p_9\}$. It may be the case that out of P_i 's choice a subset of P_i , denoted by $\hat{P}_i$, being selected finally for the implementation, i.e. $\hat{P}_i \subseteq P_i$. In this situation, motivated by [8, 39, 40], a natural utility function of any i^{th} agent will be to check that her projects to be implemented as much as possible and in that case her projects will get as much funding as possible. We can write the utility (budget utilization) of the i^{th} agent as $U_i = \sum_{p_j \in \hat{P}_i} c(p_j)$. For the last, partially funded project, $c(p_j)$ is adjusted accordingly. Given this setup, our objective is to choose a subset of projects S from the overall set of projects P based on the agents' preferences satisfying the budget limit. We also assume and take care of the fact that the weight of the ministers' votes will not be equivalent to that of the remaining agents. To achieve this goal, we take the votes of the tourists, the residents and the ministers and count them against the projects. We represent the extraction of votes for a project p_l from a vote list h as $h(p_l) = 1$ if voted for p_l , otherwise 0. Since, we have different weights for the votes of different agents, we calculate the total votes for project p_l as:

$$v(p_l) = \sum_{i=1}^n h_i^T(p_l) + \omega \times \sum_{j=1}^m h_j^M(p_l), \text{ where } l = 1, 2, \dots, q. \quad (1)$$

This, we calculate for all projects $p_l : l \in \{1, 2, \dots, q\}$. We select the projects in decreasing order of calculated votes until the budget is exhausted. The same is presented in Fig. 2. We note that there can be ties between two or more projects when the calculated votes for these projects are equal. In such cases, ties are either broken arbitrarily or some internal preference is used to prioritize one over the other.

3.2 Proposed Mechanism

The proposed mechanism, Project Selection involving Tourists, Residents and Ministers using Knapsack Voting (PSTRM-KV), takes into account the votes of the tourists, the residents and the ministers to produce the list of selected projects. Knapsack voting originally was in non-weighted setting, and here it is applied in weighted setting. The mechanism provides due weight to the votes of ministers while evaluating the results. It ensures that the total cost of the selected projects do not exceed the budget. The last project, in this case, is partially funded. The mechanism is outlined in Algorithm 1, which uses the lists of projects, their respective costs, the tourists' and the residents' votes,

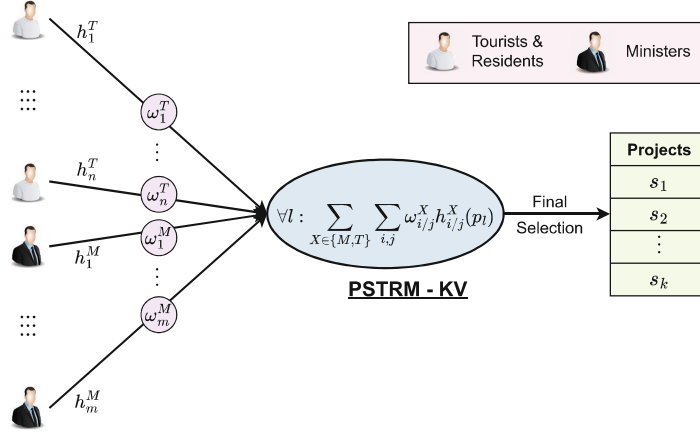


Fig. 2. Final project selection by the proposed mechanism using the input preference lists of the agents.

Algorithm 1: PSTRM-KV($\omega, T, M, P, C, \Delta$)

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1  $R, S, E \leftarrow \Phi$  // Initiate results, selected projects and expenditures.
2  $totalCost \leftarrow 0$  // Initiate total project cost.
3 foreach  $project \in P$  do // Each available project.
4   if  $project \in T$  then // Count tourist and resident votes.
5      $R[project] \leftarrow R[project] + \text{count}(T[project])$  // Count votes.
6   if  $project \in M$  then // Count minister votes.
7      $R[project] \leftarrow R[project] + \omega \times \text{count}(M[project])$  // Count votes.
8  $R \leftarrow \text{sort\_descending}(R)$  // Sort in descending order of votes.
9 foreach  $project \in R$  do // Top preferred projects.
10  if  $totalCost < \Delta$  then // If project fits budget.
11     $totalCost \leftarrow totalCost + C[project]$  // Calculate total cost.
12     $E \leftarrow E + C[project]$  // Cost of selected project.
13     $S \leftarrow S \cup project$  // Select the project.
14 return  $S, E$  // Return selected projects and expenditures.

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the ministers' votes and the total budget along with the weight factor for the ministers' votes. It calculates the votes for each available project by adding the votes of the tourists and the residents directly and that of the ministers after multiplying by the weight factor. It then sorts the projects in descending order of votes and selects the projects from the beginning of the sorted project list until the budget is just exceeded by the total selected project cost. This last selected project that just exceeds the budget is partially funded.

Table 1 shows the ordering mechanism of the PSTRM-KV mechanism before the cost based selection. Five projects, namely, 'I', 'II', 'III', 'IV' and 'V' have

been considered for the same. Also, one minister and five non-minister voters have been considered for the polling process. The weight for each minister vote has been set to $\omega = 3$. The projects are selected by the mechanism based on two criteria. Firstly, the generated weighted preference based order is to be considered for selection. Secondly, The total budget of the selected projects should not cross the available budget.

Table 1. The ordering of projects named ‘I’, ‘II’, ‘III’, ‘IV’ and ‘V’ by the PSTRM-KV mechanism using minister vote weight $\omega = 3$.

Votes Given By Ministers and Voters													
Minister 1: II I III							Voter 1: V I IV II						
Voter 2: I IV V III							Voter 3: IV V III II I						
Voter 4: I V IV III							Voter 5: III I II V						
Weighted Vote Counts for Ministers and Voters													
Ministers - I: 3, II: 3, III: 3, IV: 0, V: 0.							Voters - I: 5, II: 3, III: 4, IV: 4, V: 5.						
Final Vote Counts Before Cost Based Project Selection													
I: 8, II: 6, III: 7, IV: 4, V: 5													

4 Algorithm Analysis

The PSTRM-KV algorithm is very simple and easy to implement. The analysis of the mechanism is presented here.

Definition 1. *Strategyproofness under these settings implies that the agents cannot gain by misreporting their votes. Agents gain only when by misreporting their votes, their true preference project gets selected.*

We represent the preference list submitted by an agent A as $P_A = \{p_1, p_2, \dots, p_r\}$ for some $r < q$. We have the utility for the agent for a project p_i as $U_A^{p_i} = c(p_i)$, if p_i wins, otherwise 0. For agent A , to misreport and gain would mean that agent A doesn't report the above preference list truthfully. Instead, A submits some other preference list, say, $P' = \{p'_1, p'_2, \dots, p'_r\}$, and has utility $U'_A > U_A$.

Lemma 1. *The PSTRM-KV mechanism is strategyproof for all agents.*

Proof. Say, an agent A with true preference list as $P_A = \{p_1, p_2, \dots, p_r\}$ misreports and submits a false preference list $P' = \{p'_1, p'_2, \dots, p'_r\}$. Now, there can be two cases for the first misreported project p'_1 .

- **Case 1:** The true preference project of A gets selected (p_1). Here, the misreporting does not have any effect. Thus, the utility of the agent is $U'_A = U_A$ where the utility of project p_1 given by $U_A^{p_1} > 0$ is included.

- **Case 2:** The false preference project of A may get selected (p'_1). Now, this time A makes a loss since $U_A^{p_1} = 0$. The utility of the agent is now given by $U'_A = U_A - U_A^{p_1} = U_A - c(p_1) < U_A$.

We can argue for the rest of the projects in the same lines of logic. Thus, a misreport from any agent may result in a loss for the agent. Therefore, PSTRM-KV is strategyproof for all agents.

Lemma 2. *The PSTRM-KV mechanism produces Pareto-optimal results.*

Proof. We assume that the PSTRM-KV mechanism doesn't produce Pareto-optimal results. Let there be another mechanism, OPT, produces such results. At the start of the two mechanisms, both trivially produce Pareto-optimal results. Say, till the i^{th} iteration, both mechanisms have produced Pareto-optimal results R_i . In the $(i + 1)^{th}$ iteration, the PSTRM-KV mechanism selects the project with the maximum votes from the available ones, breaking ties in the predefined order of project priority, and produces the iteration end result of $R_{i+1}^{PSTRM-KV}$. If now, the OPT mechanism selects any other project than the one selected by PSTRM-KV, it will be a project with less than the maximum votes from the available ones, producing the iteration end result of R_{i+1}^{OPT} . Thus, at termination, the OPT mechanism will produce R^{OPT} which will not be Pareto-optimal. This contradicts our initial assumption and thus indicates that the PSTRM-KV mechanism produces Pareto-optimal results.

Lemma 3. *The PSTRM-KV mechanism runs in polynomial time.*

Proof. For this proof, consider the time complexities of the following steps. Vote counting is upper bounded as $O(nq) + O(mq) = O(nq) = O(n)$. An efficient algorithm can sort the results in descending order in $O(q) = O(1)$ complexity. Project selection has a time complexity of $O(q) = O(1)$. Here, q is the number of available projects, m is the number of ministers, n is the number of tourists and that of residents taken together, $m \ll n$ and $q \ll n$. Thus, the effective time complexity of the algorithm is capped at $O(n)$ making run in polynomial time.

Lemma 4. *The weight ω (related to minister votes) can influence the selection of project p_l .*

Proof. Let us follow project p_l through the selection process where the weight for the minister votes is given by ω . We observe what changes happen in the selection process as the weight ω changes its value:

- $\underline{\omega = \omega'}$: Say, p_l is selected for the position k in the final list in this setup. We represent the number of weighted votes received by p_l as given in Eq. (1) as $v_k(p_l)$. For the project p'_l selected for the position $k - 1$ (better position than k), we have $v_{k-1}(p'_l) > v_k(p_l)$, which gives:

$$\begin{aligned} \sum_{i=1}^n h_i^T(p'_l) + \omega' \times \sum_{j=1}^m h_j^M(p'_l) &> \sum_{i=1}^n h_i^T(p_l) + \omega' \times \sum_{j=1}^m h_j^M(p_l), \\ \implies \mathbf{A} + \omega' \times \mathbf{B} &> \mathbf{C} + \omega' \times \mathbf{D}, \end{aligned}$$

where $\mathbf{A}, \mathbf{B}, \mathbf{C}$ and $\mathbf{D}$ represent the summation terms. (2)

- $\underline{\omega = \omega''}$: Here, the weighted votes of the same project is given by $v'_k(p_l)$. For the relative positions for projects p'_l and p_l to hold true, we should have $v'_{k-1}(p'_l) > v'_k(p_l)$, which gives:

$$\begin{aligned} \sum_{i=1}^n h_i^T(p'_l) + \omega'' \times \sum_{j=1}^m h_j^M(p'_l) &> \sum_{i=1}^n h_i^T(p_l) + \omega'' \times \sum_{j=1}^m h_j^M(p_l), \\ \implies \mathbf{A} + \omega'' \times \mathbf{B} &> \mathbf{C} + \omega'' \times \mathbf{D}. \end{aligned} \quad (3)$$

Given Eq. (2), regardless of the value of ω , Eq. (3) can always be true if both $\mathbf{A} > \mathbf{C}$ and $\mathbf{B} > \mathbf{D}$ hold simultaneously (here $\mathbf{A}, \mathbf{B}, \mathbf{C}$ and $\mathbf{D}$ represent the extracted vote counts). However, we cannot claim such a relationship between these extracted vote counts apriori. Therefore, the value of $\omega \in \mathbb{R}^+$ can influence the relation between v_{k-1} and v_k . In general terms, the weight ω can influence the selection of project p_l , which concludes the proof.

5 Results

Figure 3, Fig. 4, Fig. 5 and Fig. 6 present the simulation results. In Fig. 3, 4 and Fig. 5, the order of selection for the first five chosen projects, and the weighted vote counts for the tourists, the residents, the ministers, and the result of combining them are plotted. In Fig. 3, the weight ω for the minister votes has been varied. Figure 4 presents effects of varying of the number of tourists and residents n over the selection process. Similarly, Fig. 5 presents the effects of varying the number of ministers m over the selection process. Figure 6 presents the budget utilization in percentage (%) of a random tourist or a random resident and a random minister for honest and dishonest preference submissions.

5.1 Setup

Real data has been taken from OPEN DATA BARCELONA at <https://opendata-ajuntament.barcelona.cat/data/en/dataset/est-cadaastre-carrecs-us> and has been cast to the simulation settings using Python libraries. A total of $n = \{100, 200, 300, 400, 500\}$ tourist and resident preference lists and $m = n \times \{10\%, 15\%, 20\%\}$ minister preference lists were taken as input for the simulation. A total of $q = 20$ projects (spots) with varying costs were used for the polls representing the projects with the project numbers. A budget $\Delta = \$1,000,000$ was used as the upper limit for the total cost of all the selected project costs taken together. The weight ω as in Eq. (1) was varied amongst the values $\{1, 3, 5\}$. Similarly, the numbers of tourists and residents (n) and that of the ministers (m) were also varied in the simulation.

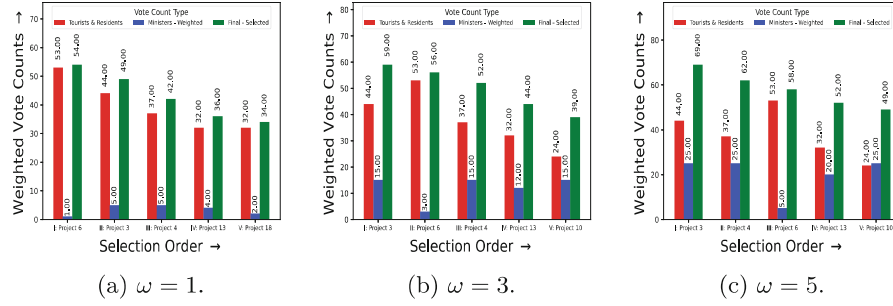


Fig. 3. Different outcomes of the selection process obtained by varying the weight ω for the votes of the ministers.

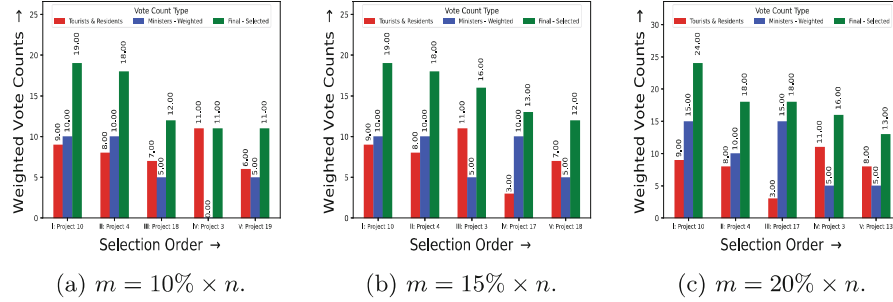


Fig. 4. Different outcomes of the selection process obtained by varying the number of ministers m with respect to the number of tourists and residents n .

5.2 Analysis

Figure 3 shows how the selection process is affected with the change in the weight ω given to the votes of the ministers. The selection process with equal weights for the tourist, the resident and the minister votes is shown in Fig. 3(a) where project number 6 is selected as the first project. As the weight changes to $\omega = 3$ as in Fig. 3(b), it is observed that project number III is selected as the first project. Similar trends are observed for the other positions when the weight is varied through $\omega = \{1, 3, 5\}$. With the variation of ω , the selection of projects shuffles through several sets distinct from the one with $\omega = 1$, i.e. equal weights for the tourist, the resident and the minister votes. Ties between projects have been broken by assuming a higher priority for a project with a lower project number amongst the tied projects. Figure 4 and Fig. 5 shows how varying the number of tourists and residents (n) and that of the ministers (m) with respect to each other affect the selection process. While the variation of m affects the selection of the projects lower in the selection order as in Fig. 4, the variation of n affects the selection order from the very first position as in Fig. 5. The effect of change of the number of ministers m depend heavily on the weight ω selected for their votes. The effect increases with increasing values of ω . The number of tourists

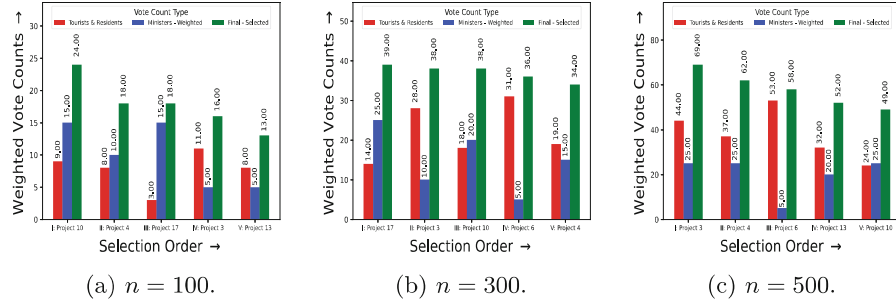


Fig. 5. Different outcomes of the selection process obtained by varying the number of tourists and residents n with respect to the number of ministers m .

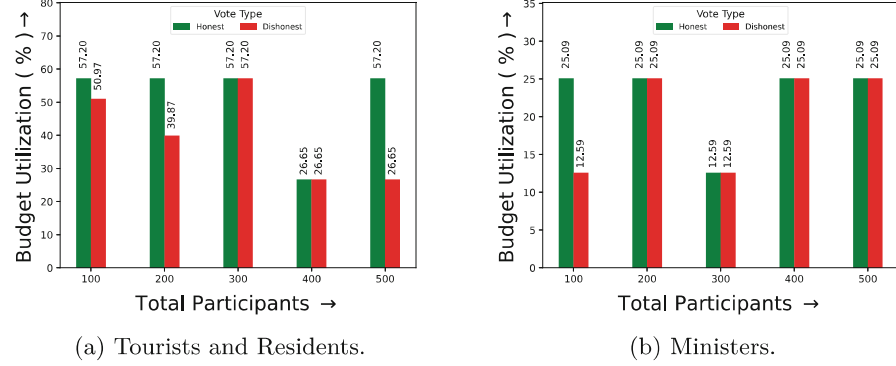


Fig. 6. Budget utilization by random agents when submitting honest and dishonest preference lists in the system.

and residents n has a profound effect on the process because of their larger numbers when compared to that of the ministers. Figure 6 shows the budget utilization of a random tourist or resident and a random minister while they provide honest and dishonest preferences to the system. It is seen that although sometimes there is no loss of an agent in terms of budget utilization, there is never any gain either. Since, the number of ministers is almost always less than that of the rest of the agents, there is a lower chance that a minister loses due to misreporting. However, as the weight of each vote of ministers increase, this chance increases too.

6 Discussion

From the simulation results the following points can be inferred:

- The weighted vote of ministers will definitely have an effect on the outcome of the selection process for the projects to be undertaken.

- The weight per vote for the ministers plays an important role in the selection process as different weights can turn the scales differently for the projects.
- Strategyproofness makes the PSTRM-KV mechanism very attractive for both the parties, namely, the tourists, the residents and the ministers.
- The fact that the PSTRM-KV mechanism produces Pareto-optimal results implies that the tourists, the residents and the ministers both can depend on the optimal nature of the selected list of projects.

We have introduced ω in Sect. 3.1 to capture the difference in preference weights of the ministers (managers) and that of the tourists and residents. Here, ω can be varied to achieve different effects (thereby, outcomes) on the project selection process. Simulation has also been carried out using different values of ω (weight of ministers' votes) to show the change in outcome due to different weight selection. The simulation results showing the effect of varying ω on the outcome of project selection is presented in Fig. 3.

Here, we are accepting the votes from two groups. The first one are the tourists and the residents of the place. The second one is the ministers (they could be interpreted as managers of any organization). This paper gives insights on how to take votes from both of them so that a tourist spot could be determined in such a way that due weightage is given to the respective groups (that is, including the managerial group). So here, managerial roles are also important in selection of the spot.

7 Conclusion

Tourism is a great source of national income and the Government is well aware of the same. That is why the Government wants to know about the projects that the tourists and the residents want to be built in addition to what projects the ministers want there. The challenge is to take into account the preferences of the tourists, the residents and the ministers to create a preference list following which the projects will finally be built. In addition, during the final preference list creation, the different weights to be used for the ministers' preferences is to be accommodated into the algorithm.

The proposed mechanism, namely, PSTRM-KV, takes the votes of the tourists, the residents and the ministers and produces a final list of selected projects to be undertaken by the Government. The mechanism accounts for the variable weights given to the votes of the ministers while calculating the resulting selection list. This encourages tourists and residents to participate in the process actively. In addition, it retains the much required ministry control over the process in the way of the preferences from the ministers. While the tourists, the residents and the ministers may misreport their votes, the strategyproofness property of the mechanism ensures that they are not motivated to do so and if they at all do, they will not gain by misreporting their preferences. Most importantly, the weight parameter that determines the computational contribution of the ministers' votes in the selection process, can be tweaked to suit the needs of

the land thereby making the algorithm suitable for a broad spectrum of practical settings.

Frequent visitors of the tourist place may be tracked and they may be prompted to participate in the voting process whenever it is held. Their votes may also be given non trivial weight in the selection process. This direction could be potential future works.

Funding. No funding was received for conducting this study.

Competing Interests. The authors have no competing interests to declare that are relevant to the content of this article.

Availability of Data and Code. Real data has been taken from OPEN DATA BARCELONA at <https://opendata-ajuntament.barcelona.cat/data/en/dataset/est-cadastre-carrecs-us> and has been cast to the simulation settings using Python libraries. The source data, the cast data, the Python codes and the generated graphs are available in GitHub at <https://github.com/Surja-Sanyal/Crowdsourced-Project-Selection-in-Tourist-Spots.git>.

Author Contributions. All authors contributed equally to this work.

Ethics Approval. Not applicable.

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